

Safety Data Sheet

1 Product and company identification

Product name	O-MACK No.200 CLEAR
Product explanation	Kind Specially modified silicate resin figure paint Use Paint for Metals
Manufacturer	Ohashi Chemical Industries Ltd.
Address	6-21, Hatakeda-cho, Ibaraki, Osaka, Japan
Section	Division of Quality Certification OKAYAMA FACTORY
Telephone number	+81-868-38-4131
Fax number	+81-868-38-1788
Emergency telephone number	+81-868-38-4131

2 Hazards identification

GHS label element
Symbol or pictogram



Signal word

Danger

GHS classification	Category	Hazard statement
Physical hazards		
Flammable liquids	2	Highly flammable liquid and vapour
Health hazards		
Acute toxicity Oral	4	Harmful if swallowed
Skin	3	Toxic in contact with skin
Inhalation/Gases		Not available classification
Inhalation/Vapours	3	Toxic if inhaled
Inhalation/Dust,Mist	-	Others
Skin corrosion/irritation	2	Causes skin irritation
Serious eye damage/irritation	2	Causes serious eye irritation
Respiratory sensitization	-	Others
Skin sensitization	1	May cause an allergic skin reaction
Germ cell mutagenicity	2	Suspected of causing genetic defects
Carcinogenicity	1	May cause cancer
Toxic to reproduction	1	May damage fertility or the unborn child
Specific target organ system toxicity (single exposure)	1	Causes damage to organs [Liver, Kidney, Blood, Body as a whole, Respiratory, Central nerve system]
	3	May cause respiratory irritation; or May cause drowsiness or dizziness [Periodontal immune system]
Specific target organ system toxicity (repeated exposure)	1	Causes damage to organs through prolonged or repeated exposure [Bone, Blood, Central nerve system]
	2	May cause damage to organs through prolonged or repeated exposure [Liver, Kidney, Hearing, Spleen, Respiratory, Immune system]
		Not available classification
Aspiration hazard		
Environment hazards		
Aquatic toxicity (acute)	3	Harmful to aquatic life
Aquatic toxicity (chronic)	3	Harmful to aquatic life with long lasting effects
Hazardous to the ozone layer	*	Classification not possible

Precautionary statement

Preventive measures

- Obtain special instructions before use.
- Do not handle until all safety precautions have been read and understood.
- Keep away from heat/sparks/open flames/hot surfaces. – No smoking.
- Keep container tightly closed.
- Ground/Bond container and receiving equipment.
- Use explosion-proof electrical/ventilating/lighting equipment.
- Use only non-sparking tools.
- Take precautionary measures against static discharge.
- Do not breathe dust/fume/gas/mist/vapours/spray.
- Avoid breathing dust/fume/gas/mist/vapours/spray.
- Wash hands well after handling.
- Do not eat, drink or smoke when using this product.
- Use only outdoors or in a well-ventilated area.
- Contaminated work clothing should not be allowed out of the workplace.
- Avoid release to the environment.
- Wear protective gloves/protective clothing/eye protection/face protection.

First-aid measure

- IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.
- IF ON SKIN: Wash with plenty of soap and water.
- IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
- IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
- IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
- IF exposed: Call a doctor.
- IF exposed or concerned: Get medical advice/attention.
- Call a POISON CENTER or doctor/physician.
- Call a POISON CENTER or doctor/physician if you feel unwell.
- Get medical advice/attention if you feel unwell.
- Measures are necessary especially.
- Rinse mouth.
- If skin irritation occurs: Get medical advice/attention.
- If skin irritation or rash occurs: Get medical advice/attention.
- If eye irritation persists: Get medical advice/attention.
- Take off immediately all contaminated clothing and wash it before reuse.
- Take off contaminated clothing and wash it before reuse.
- In the case of fire: Use appropriate extinguishant.

Storage

- Store in a well-ventilated place. Keep container tightly closed.
- Store in a well-ventilated place. Keep cool.
- Store locked up.

Waste

- Dispose of contents/container according to the rule of each country.

3 Composition / information on ingredients

Substance / Mixture : Mixture

Chemical name	CASNo.	CONC.	Remarks
Poly[2-(chloromethyl)oxirane-alt-4,4'-(propane-2,2-diyl)diphenol]	25068-38-6	0.7%	
Silicon dioxide	7631-86-9	7.7%	
Xylene	1330-20-7	0.6%	
Ethylbenzene	100-41-4	0.5%	
Methanol	67-56-1	0.1%	
Propan-2-ol	67-63-0	17.6%	
1-Butanol	71-36-3	1.6%	
Diacetonealcohol	123-42-2	15.6%	
Butyl acetate	123-86-4	1.8%	
2-Butoxyethanol	111-76-2	15.6%	
Methyl isobutyl ketone	108-10-1	0.3%	
Cyclohexanone	108-94-1	24.4%	
Formaldehyde	50-00-0	0.2%	

4 First-aid measure

Inhalation

- If a vapor or gas is inhaled in large amounts, immediately move to a clean place for a breath of fresh air and take a rest, keeping yourself warm. In case of suffocation, artificial respiration should be practiced. Care should be taken so that anything thrown up may not be swallowed. Medical treatment should be received immediately.
- If you have inhaled a vapor or gas and feel ill, move to a clean place for a breath of fresh air, keep quiet, and take medical treatment.

Skin contact

- Immediately wipe the product off your skin with a cloth.
- Wash away the matter thoroughly with water and soap or a with skin cleaning detergent. Do not use a solvent or thinner.
- If the contacted skin changes in appearance or gives you pain, consult a doctor.
- Immediately to remove all contaminated clothing.
- To remove all contaminated clothing.

Eye contact

- Immediately rinse your eyes under clean running water for at least 15 minutes. Even the back sides of the eyelids should be washed thoroughly.
- See a doctor as soon as possible.
- Medical treatment should be received immediately.

Ingestion

- If the product is swallowed by mistake, lie quietly and immediately take medical treatment.
- Care should be taken so that anything thrown up may not be swallowed.
- The person who swallowed the product should not be induced to vomit without directions from a doctor.

Protection of first-aiders

- Appropriate protective equipment (safety glasses, masks, and gloves) to wear. Provide ventilation.

5 Fire-fighting measures

Extinguishing media

- Reinforced liquid (fog), Bubble, Carbon dioxide, halogenide, Powder (the use of phosphates), Powder (the use of hydrogencarbonates), Dry sand

NOT suitable extinguishing media

- Straight stream water, Water fog, Reinforced liquid (straight stream)

Protection of fire-fighters

- Wear appropriate protectors (such as heat-resistant clothes).
- Move any combustible materials away from the fire.
- Use a fire extinguishing agent of the specified type.
- Pour water on any heat-exposed sealed containers to cool them.
- Fight the fire from the windward side.

6 Accidental release measures

Personal precautions, protective equipment and emergency measures

- Wear appropriate protectors (such as gloves, protective mask, an apron, goggles)
- Remove any nearby ignition or heat sources and combustible materials.
- Keep an appropriate fire extinguisher at the ready to guard against ignition.
- To prevent secondary disasters by non-participants to hold the place and the area off limits to its location.
- Provide sufficient ventilation for indoors
- As much as possible work at a windward location for outdoors.

Environmental precautions

- Be careful so that the leakage may not be discharged into a river or other places to damage the environment.

Method and materials for containing

- Put the leakage into an gastight container and move it to a safe place.
- Dispose of any adherends and wastes in accordance with the related laws and regulations.
- For recovery, use tools made of materials such as would not produce sparks on impact or due to static electricity.
- Dry sand, earth, we recover what is absorbed into other non-combustible. The outflow of mass enclosed in the embankment to prevent leakage.

7 Handling and storage

Handling

- Handle the product in a well-ventilated place.
- Completely seal the container after closing it.
- Do not use any heat sources or spark-producing devices nearby.
- To guard against static electricity, short the equipment to ground and use only explosion proof (increased safety type) electric apparatus.
- Use a spark-proof tool.
- Immerse the used rags and spray cans sin water until they are disposed of.
- Note that this product is flammable if a high temperature.
- It touches the skin, the mucous membrane or clothes, and so as not to catch one's eye, the protective equipment is worn.
- After handling, neither the hand nor the face, etc. wash well, and bring the pollution protection tool such as gloves in to the rest room etc.
- When handling the product in an enclosed space, furnish an appropriate local exhauster and wear an appropriate protectors.
- During handling and use of type antistatic work clothing and shoes.
- A person who has acquired an allergy should not handle the product.

Storage

- Avoid storing the product in a place exposed to direct sunlight.
- Store the product in a well-ventilated place.
- Store the product in a place where there are no heat sources nearby.
- Keep it locked to prevent theft.
- Keep away from children.
- Keep away from acids and alkaline.

8 Exposure controls / personal protection

Control parameters, standard control concentration

Chemical name	Standard control	ACGIH TWA ppm	Remarks
Poly[2-(chloromethyl)oxirane-alt-4,4'-(propane-2,2-diyl)diphenol]	—	—	
Silicon dioxide	—	—	
Xylene	50 ppm	100 ppm	
Ethylbenzene	20 ppm	20 ppm	
Methanol	200 ppm	200 ppm	
Propan-2-ol	200 ppm	200 ppm	
1-Butanol	25 ppm	20 ppm	
Diacetonealcohol	—	50 ppm	
Butyl acetate	150 ppm	150 ppm	
2-Butoxyethanol	25 ppm	20 ppm	
Methyl isobutyl ketone	20 ppm	20 ppm	
Cyclohexanone	20 ppm	20 ppm	
Formaldehyde	—	—	

Engineering measures

- Use equipment of the explosion proof type.
- With the exhaust system to prevent vapor accumulation.
- Short any liquid conveying, dip-up, an stirring devices to ground.
- Do not use any heat sources or spark-producing devices nearby.
- When handling the product indoors, use equipment such as would protect the operator from direct exposure to it or allow him to avoid exposure as by a local exhaust.
- When handling the product in an enclosed space such as the inside of a tank, furnish a ventilator such as can circulate fresh air through the entire workplace.

Respiratory protection

- Wear a gas mask for organic gases.
- When handling the product in an enclosed space, wear a full-face air-supplied respirator.

Hand protection

- Wear gloves made of materials which do not allow penetration of organic solvents of chemicals.

Eye protection

- Wear safety goggle.

Skin and body protection

- Wear clothes such as would protect the skin from direct exposure. It is desirable that the clothes be made of materials which do not allow penetration of chemicals.

Others

- For electrostatic coating, wear current-carrying shoes.

9 Physical and chemical properties

Form	: Liquids
Color	: Colorless and transparent
Odor	: Organic solvent
PH	: Not available
Melting point	: Not available
Freezing point	: Not available
Boiling point	: 83.0°C(Reference value)
Boiling range	: 83.0~171.0°C(Reference value)
Flash point	: 12°C
Lower explosive limit	: 1.1%(Reference value)
Upper explosive limit	: 12.7%(Reference value)
Vapour pressure	: 4400Pa/20°C(Reference value)
Specific gravity	: Not available
Solubility	: Not available
Octanol / water partition coefficient	: Not available
Ignition point	: 240°C(Reference value)

10 Stability and reactivity

Stability

- No reaction occurs under normal conditions.

Condition to avoid

- Heating the product may cause a dangerous reaction.

Incompatible products

- This product could cause a dangerous reaction in contact with strong oxidants.

Hazardous decomposition products

- Decomposition may release toxic gases.

Other danger information

- Vapor is flammable, it may cause an explosion.

11 Toxicological information

Acute toxicity

Category		ATEmix		
Oral	Category4	LD50	1156.9	mg/kg
Skin	Category3	LD50	851.8	mg/kg
Inhalation/Gases	x	LC50		ppmV
Inhalation/Vapours	Category3	LC50	1324.6	ppm
Inhalation/Dust,Mist	-	LC50	32.1	mg/l

- : Others. x : Not available classification. * : Classification not possible.

Chemical name	Acute toxicity					Skin corrosion /irritation	Eye damage /irritation
	Oral	Skin	Inhalation /Gases	Inhalation /Vapours	Inhalation /Dust,Mist		
Poly[2-(chloromethyl)oxirane-alt-4,4'-(propane-2,2-diyl)diphenol]	-	*	x	*	*	Category2	Category2B
Silicon dioxide	-	-	x	x	*	-	Category2
Xylene	-	-	x	-	*	Category2	Category2A
Ethylbenzene	-	-	x	Category4	*	-	Category2B
Methanol	Category4	-	x	-	*	*	Category2
Propan-2-ol	-	-	x	-	*	-	Category2
1-Butanol	Category4	-	x	-	*	Category2	Category2A
Diacetonealcohol	-	-	x	*	*	Category2	Category2A
Butyl acetate	-	-	x	x	*	-	Category2B
2-Butoxyethanol	Category4	Category3	x	Category2	*	Category2	Category2A
Methyl isobutyl ketone	-	-	x	Category3	*	-	Category2B
Cyclohexanone	Category4	Category3	x	Category3	-	Category2	Category2A
Formaldehyde	Category4	Category3	Category2	*	*	Category2	Category2A

Chemical name	Respiratory sensitization	Skin sensitization	Germ cell mutagenicity	Carcinogenicity	Toxic to reproduction	Aspiration hazard	CONC.
Poly[2-(chloromethyl)oxirane-alt-4,4'-(propane-2,2-diyl)diphenol]	*	Category1	-	*	-	*	0.7%
Silicon dioxide	*	*	*	Category1A	*	*	7.7%
Xylene	*	*	-	-	Category1B	*	0.6%
Ethylbenzene	*	*	-	Category2	Category1B	Category1	0.5%
Methanol	*	-	-	*	Category1B	*	0.1%
Propan-2-ol	*	*	*	*	Category2	*	17.6%
1-Butanol	*	*	-	-	-	*	1.6%
Diacetonealcohol	*	*	*	*	Category2	*	15.6%
Butyl acetate	*	-	*	*	*	*	1.8%
2-Butoxyethanol	*	-	-	-	Category2	*	15.6%
Methyl isobutyl ketone	*	-	-	Category2	-	*	0.3%
Cyclohexanone	*	Category1	Category2	-	Category2	*	24.4%
Formaldehyde	Category1	Category1	Category2	Category1A	*	*	0.2%

Chemical name	Specific target organ system toxicity	
	Single exposure	Repeated exposure
Poly[2-(chloromethyl)oxirane-alt-4,4'-(propane-2,2-diyl)diphenol]	Classification not possible	Classification not possible
Silicon dioxide	Category 3 (Respiratory tract pungency)	Category 1 (Kidney, Respiratory, Immune system)
Xylene	Category 1 (Liver, Kidney, Respiratory, Central nerve system) Category 3 (Periodontal immune system)	Category 1 (Nerve system, Respiratory)
Ethylbenzene	Category 2 (Central nerve system) Category 3 (Respiratory tract pungency)	Classification not possible
Methanol	Category 1 (Body as a whole, Sight-organ, Central nerve system) Category 3 (Periodontal immune system)	Category 1 (Sight-organ, Central nerve system)
Propan-2-ol	Category 1 (Body as a whole, Central nerve system) Category 3 (Respiratory tract pungency)	Category 1 (Blood) Category 2 (Liver, Spleen, Respiratory)
1-Butanol	Category 3 (Respiratory tract pungency, Periodontal immune system)	Category 1 (Hearing, Central nerve system)
Diacetonealcohol	Category 2 (Liver, Blood) Category 3 (Respiratory tract pungency, Periodontal immune system)	Classification not possible
Butyl acetate	Category 2 (Respiratory, Central nerve system)	Classification not possible
2-Butoxyethanol	Category 1 (Liver, Kidney, Blood, Central nerve system) Category 3 (Respiratory tract pungency)	Category 2 (Blood)
Methyl isobutyl ketone	Category 3 (Respiratory tract pungency, Periodontal immune system)	Category 1 (Nerve system)
Cyclohexanone	Category 1 (Respiratory) Category 2 (Central nerve system)Category 3 (Periodontal immune system)	Category 1 (Bone, Central nerve system)
Formaldehyde	Category 1 (Nerve system, Respiratory)	Category 1 (Respiratory, Central nerve system)

12 Ecological information - : Others. x : Not available classification. * : Classification not possible.

Chemical name	Aquatic toxicity		Hazardous to the ozone layer
	Acute	Chronic	
Poly[2-(chloromethyl)oxirane-alt-4,4'-(propane-2,2-diyl)diphenol]	Category1	Category1	*
Silicon dioxide	*	*	*
Xylene	Category2	Category2	*
Ethylbenzene	Category1	-	*
Methanol	-	-	*
Propan-2-ol	-	-	*
1-Butanol	-	-	*
Diacetonealcohol	-	-	*
Butyl acetate	Category3	-	*
2-Butoxyethanol	-	-	*
Methyl isobutyl ketone	-	-	*
Cyclohexanone	-	-	*
Formaldehyde	Category2	-	*

General instructions

- When the product leaks or when it is to be disposed of, special care should be taken because it may damage the environment.
- In particular, the product or its wash water should never be discharged direct to the ground or to a river or a drain ditch.

Other

Ecotoxicity

- Not available

Persistence/degradability

- Not available

Bioaccumulation

- Not available

Mobility in soil

- Not available

13 Disposal considerations

Waste from residues

- The waste should be disposed of under a contract with an authorized industrial waste disposal company.
- Container and apparatus wash water should not be discharged direct to the ground or to a drain ditch.
- The industrial waste disposal trader who obtains permission and the consignment contract are done and it processes it because it corresponds to the special management industry waste (wasted oil).
- Follow the relevant laws and local government standards for waste disposal.
- The wastes producing from process of water refining and of incineration should be disposed of in accordance with governmental laws and environmental control regulations or asked to dispose with licensed special company.
- It disposes of an empty container after contents are completely removed.

Contaminated packaging

- It disposes of an empty container after contents are completely removed.
- The waste should be disposed of under a contract with an authorized industrial waste disposal company.

14 Transport information

UN numbers : 1263
UN name : Paint related Materials including paint, flammable
UN class : Class 3 Flammable Liquids
Container grade : II
Guideline Number : 128

Special safety measures

Common

- Observe the handling and storage precautions.
- Check the container for leakage, load it with care so that it will not topple over, fall down, or sustain damage, and take adequate measures to prevent load shifting.

Land transportation

- The product should be transported in accordance with Fire Service Law, the industrial Safety and Health Law, and the Poisonous and Deleterious Substances Control Law.
- The shipper issues transportation instructions (Yellow-Card) to the transporter.

Marine transportation

- Maritime transportation should be in compliance with 「Ship Safety Act」. Air transportation should be in compliance with the Aviation Act.

Air transport

- The product should be transported in accordance with the Aviation Act.

15 Regulatory information

- IMPORTANT: Follow all regulation in your country.

16 Other information

Main references cited

- Japanese Paint Manufacturers Association editing "Database for paint".
- Handbook of solvent.
- Chemical article (Chemical Daily).
- "Guidebook for SDS-label making" which Japan Paint Manufacturers Association edited for GHS (Revised 2nd Edition).
- ACGIH: Threshold Limit Values for Chemical substances and Physical Agents.
- NITE (product evaluation technology base mechanism) offer GHS classification result DB.

Limitation of the responsibility.

- Safety data sheet what is currently available materials, information, and are prepared in accordance with our data may be revised or amended by new knowledge of laws.
- When handling this product, please take security measures adapted to the reality and responsibility of the user in reference to the contents listed.
- This data sheet is no guarantee of safety and quality.